

Plans Keep Secrets: External Outlines Stop Involuntary Leakage, and the Leaked Secret Lives in the Residual Stream

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Abstract

Language models told to keep a secret while writing tend to leak it. A story written while holding the secret word “cactus” drifts towards dust and heat, and a second model can often tell which of two stories was written with which secret [10]. We ask whether giving the writer an explicit plan prevents this, and what the leak is made of mechanistically. If planning and prose generation are entangled, a writer that makes up content as it goes may keep reaching for what it holds in mind, and an outline that fixes the content should remove that pressure. With Gemma-3-12B-it as the writer and a paired two-alternative forced-choice guesser, the plain “keep it secret” condition leaks (GPT-4.1 guesser accuracy 0.70; chance 0.50). Following a secret-free outline removes the leak (short outline 0.54, detailed outline 0.48). Length-matched irrelevant text does not (0.71), so the effect is not dilution. A plan written by the secret-holder itself does not help: the plan leaks (0.80), and so does the story that follows (0.76). Holding the story text fixed and varying only the secret in the prompt shows that the secret is linearly decodable from story-token activations (up to 100%, 15-way). The logit lens ranks the secret word first among 15 candidates at middle layers but suppresses it at the output. The secret becomes more active as the story progresses, and per-story signal strength predicts per-story leakage (Spearman $\rho = 0.46$). Outlines leave the secret decodable but cut its signal about 4 $\times$, while irrelevant text leaves it unchanged. Swapping texts and contexts shows that the plan in context, not the text it produced, is what quiets the secret. Mean-ablating a single secret direction per layer during generation lowers leakage (0.70 $\rightarrow$ 0.59) without hurting story quality, while matched controls do not; adding the direction to a model with no secret induces leakage. Twist endings leak only marginally under a “do not foreshadow” instruction (0.53, vs. 0.69 when asked to foreshadow). We also show that naive zero-ablation *amplifies* the concept, because of Gemma’s massive activation dimensions.

1 Introduction

When language models write fiction, they tend to telegraph what is coming. One explanation is that *planning and prose generation are entangled*: a model that “knows” the twist must keep the twist active in its internal state in order to steer towards it, and that activity bleeds into word choice, imagery, and emphasis. Holtzman and West [10] turned this intuition into a measurable phenomenon. A writer model receives a secret word and an instruction not to mention or hint at it, then writes a story; a guesser model, shown two stories written with two different secrets, picks which one belongs to a given secret. Several frontier models are reliably detected (70–79%), even though they never mention the word.

That work is behavioural, and two questions it leaves open motivate this paper. First, the original idea behind it: **does conditioning on an explicit plan reduce leakage?** If the model leaks because it has to make up content on the fly while holding the secret, then an outline that pins down the content should leave little room for the secret to seep in. An outline also simply adds context, which might dilute the secret’s salience, so the comparison needs a length-matched irrelevant-context control. Second, **what is the mechanism?** Is the secret concept actually represented in the residual stream throughout generation? Does its strength predict leakage? Does removing it remove the leak?

We study these questions with Gemma-3-12B-it [8], the smallest model reported to leak strongly [10], so that we can intervene on activations. Our contributions:

- **Plans stop leakage, and it is not dilution.** Following a secret-free outline, whether short ($\sim$ 100 words) or detailed ($\sim$ 540 words), drives guesser accuracy from 0.70 to chance, whereas length-matched irrelevant context leaves leakage unchanged. A plan composed by the writer *itself* while holding the secret does not help, because the leak moves into the plan.
- **Plot-level secrets.** When the writer knows a twist ending and writes the opening under a “do not foreshadow” instruction, leakage is small (0.53) compared with an explicit request to foreshadow (0.69). An outline brings it to chance.

- **The secret persists internally, and leakage tracks it.** With the text held fixed, the held secret is decodable from story tokens at up to 100% (15-way). It is the model’s top candidate under the logit lens at middle layers but suppressed at the output. It grows more active over the story, and its per-story strength predicts leakage. Outlines weaken this representation about $4\times$ and stop it from growing, without erasing it. A cross-context test shows that this is caused by the plan in context rather than by the text it produced.
- **Causal evidence and a pitfall.** Mean-ablating the secret direction during generation reduces leakage at no cost to quality, whereas other-word and random ablations do not. Injecting the direction into a model with no secret creates leakage. Naive zero-projection ablation instead *amplifies* the concept (“coastal coastal coastal...” for *lighthouse*), because Gemma’s massive activation dimension dominates the projection.

2 Related work

Involuntary leakage. Holtzman and West [10] introduced the secret-word 2AFC paradigm and showed leakage across frontier models, its suppression by an “actively hide” instruction, its absence in short jokes, and its redirection by decoy words. They also reported a size threshold (no leakage for Llama-3.2-3B/3.1-8B, strong leakage for Gemma-3-12B/27B). *Semantic leakage* [9] is the related tendency of a concept in the prompt to colour generations in semantically associated ways, e.g., “yellow” leading to “school bus”. Ironic negation [13] parallels the human white-bear effect [19]: an instruction not to mention X can increase X ’s accessibility. Our mechanistic results are consistent with this account. The secret is strongly active internally, and it is suppressed only at the final layers.

Secrets and private state. Contextual-integrity work asks whether models disclose private information inappropriately [14]. Baldelli et al. [2] prove that secrecy plus consistency is impossible without private working memory, and Luo et al. [12] find that models lack stable implicit commitments in 20-questions games. In our setting the secret is in the context, and the question is whether it seeps into the output.

Eliciting secrets from activations. Taboo model organisms fine-tune a secret word into a model that hints at it, and recover the word with logit lens and SAEs [4, 5]. We instead study an *instructed*, in-context secret and *involuntary* leakage into unrelated text, and we use counterfactual teacher forcing to separate the internal representation from the content of the text.

Planning in LMs and in story generation. Hidden states encode information about future tokens and response attributes [6, 16, 20], and models plan rhyme targets ahead in poems [11]. Outline-conditioned generation is an established way to improve coherence [7, 22, 23], and it has been used for suspense [21]. As far as we know, no prior work has tested whether an outline reduces involuntary leakage.

Directional interventions. We use difference-of-means directions and directional ablation in the spirit of Ardit et al. [1], activation addition [18], and the logit lens [3, 15].

3 Experimental setup

3.1 Writer, secrets and prompts

The writer is **Gemma-3-12B-it** (bf16, HuggingFace `transformers`; temperature 1.0, top- p 0.95, top- k 64, max 800 new tokens). We use the 15 secret words of Holtzman and West [10] (umbrella, lighthouse, violin, cactus, telescope, justice, patience, entropy, nostalgia, freedom, bracket, Tuesday, copper, margin, invoice) and 20 neutral story premises (e.g., “two strangers who share a long train journey”). Set A (10 premises) is used for the behavioural conditions and for estimating directions. Set B (10 held-out premises) is used for interventions. Every (premise, secret) pair yields one story, giving 150 stories per condition. The base prompt is: *Your secret word is “w”. Keep it secret: do not mention the word and do not hint at it in any way. The story must not reveal anything about the secret word. Write a short story (about 450 words) about [premise].*

Word-level conditions. (1) **Secret** (the positive anchor): the base prompt alone. (2) **+ Short outline**: the prompt adds “Follow this outline closely, beat by beat” and a 5-bullet outline (~ 100 words). (3) **+ Detailed outline**: a 12-beat outline (~ 540 words). Outlines are written by the *same model without any secret*, are shared by all 15 secrets for a premise, and are filtered to contain no secret word. (4) **+ Irrelevant text**: an encyclopedia passage (bees, sourdough, ...) truncated to the length of the detailed outline, introduced as an unrelated part of the prompt template. This controls for added context and dilution. (5) **+ Self-written plan**: the model first writes its own outline under OUTLINE, while holding the secret, and then the story. We score the story part, and separately the plan. (6) **+ Decoy**: the model is told to think about an unrelated decoy word (the known mitigation, included as a yardstick).

3.2 Measuring leakage: paired 2AFC

For premise p and secret w_i , we pair story $s_{p,i}$ with story $s_{p,j}$ written for the *same premise, same condition* but secret w_j , where $j = (i + 1 + (p \bmod 14)) \bmod 15$. The guesser is told that one writer had the secret “ w_i ” and is asked which story it was. Because both stories share the premise, the condition, and (where present) the outline, any systematic difference reflects the secret. Each question is asked in both presentation orders, and the guesser’s choice is read off its next-token probabilities over $\{1,2\}$. Our score per question is the order-averaged probability of the correct answer (“accuracy”; chance = 0.5); the order-averaged binary accuracy gives nearly identical numbers. We use two guessers. **GPT-4.1** via OpenRouter, temperature 0 with logprobs, is our primary guesser. **Qwen3-14B** [17], run locally in non-thinking mode, serves as a fully reproducible second guesser. Both guessers are strongly position-biased: GPT-4.1 chose the first story 71% of the time on the baseline. Averaging over the two orders removes this bias. Confidence intervals come from a *crossed* bootstrap that resamples premises and secrets independently (4,000 replicates). This is conservative, because word-level heterogeneity is large. Condition contrasts are computed on matched questions, with the same crossed bootstrap applied to paired differences.

3.3 Plot-level secrets

Claude Sonnet 4.5 generated 50 premises, each with two mutually exclusive twist endings that are equally plausible from the premise, plus a neutral 6-beat outline for the opening that is compatible with both twists. The writer is given the premise and one twist and writes only the *opening* (~ 400 words); the twist would appear in a later part it does not write. The conditions are: **hide** (do not reveal or foreshadow the twist), **hide + outline**, **hide + irrelevant text** (length-matched to the outline), **foreshadow** (“subtly foreshadow the twist”, a positive control for guesser sensitivity), and **no twist** (control). Each premise $\times$ twist is sampled twice, giving 200 openings per condition (100 for no twist). The guesser sees the premise, both endings, and the opening, and picks the ending the writer was setting up (both orders). Because every premise appears with both twists, any premise-level prior favouring one twist cancels. Indeed, with no twist given, GPT-4.1 picked twist A 52% of the time and Qwen3 59%. CIs come from a premise-cluster bootstrap.

3.4 Mechanism: counterfactual teacher forcing

To separate the model’s *internal state* from the *content* of what it wrote, we take each generated story x (written with secret w) and re-run the model on $[\text{prompt}(v); x]$ for all 15 secrets v , keeping the condition, the outline, and the story tokens fixed. The text is identical across the 15 runs, so any variation of story-token activations with v reflects the held secret rather than the text. For each layer ℓ , we average the residual stream (output of block ℓ) over story tokens and centre it across the 15 variants, giving $z_{x,v,\ell}$. The **secret direction** $\hat{d}_{w,\ell}$ is the normalised mean of $z_{x,w,\ell}$ over set-A baseline stories, estimated leave-one-premise-out when evaluated on set A. We report: (a) **decodability**, i.e., whether $\arg \max_u \langle z_{x,w,\ell}, \hat{d}_{u,\ell} \rangle = w$ (15-way, chance 6.7%); (b) the **secret signal**, $\langle z_{x,w,\ell}, \hat{d}_{w,\ell} \rangle$ minus the mean over other directions, per story and per token; (c) a **logit lens** readout: the final norm and unembedding applied to story-token residuals, restricted to the first tokens of the 15 secret words and centred across variants; and (d) the Spearman correlation between per-story signal and that story’s guesser accuracy.

3.5 Interventions

Mean-ablation. During generation on held-out set-B premises, at every layer and every *generated* token, we clamp the residual’s projection onto a subspace U_ℓ to its counterfactual mean, i.e., the value expected when the model holds

Table 1: Word-level leakage (Gemma-3-12B writer, 150 stories per condition; chance = 0.50). Accuracy is the order-averaged probability that the guesser picks the true secret-holder. CIs come from a crossed bootstrap over premises $\times$ secrets, and Δ is the paired difference from the *secret, no plan* condition. “Mention” is the share of stories containing the secret word. The last two columns come from the counterfactual activation analysis at layer 32 (Section 4.3): 15-way decoding accuracy of the held secret, and mean held-secret signal.

Condition	GPT-4.1 guesser		Qwen3-14B guesser		mention %	words	internal (L32)	
	acc. [95% CI]	Δ vs. secret (p)	acc. [95% CI]	Δ (p)			decode	signal
Secret, no plan	0.70 [0.59, 0.80]	–	0.64 [0.54, 0.74]	–	0.0	433	0.99	318
+ irrelevant text (len.-matched)	0.71 [0.58, 0.82]	+0.01 (0.904)	0.66 [0.56, 0.76]	+0.02 (0.677)	1.3	360	0.99	332
+ short outline (self, secret-free)	0.54 [0.47, 0.61]	-0.16 (0.004)	0.51 [0.44, 0.58]	-0.13 (0.019)	0.0	463	0.93	128
+ detailed outline (self, secret-free)	0.48 [0.42, 0.54]	-0.22 (0.002)	0.51 [0.46, 0.55]	-0.14 (0.028)	0.0	530	0.83	75
+ self-written plan (story part)	0.76 [0.63, 0.87]	+0.06 (0.221)	0.68 [0.58, 0.78]	+0.04 (0.440)	0.0	439	0.96	167
+ decoy word	0.50 [0.40, 0.60]	-0.20 (0.017)	0.48 [0.38, 0.58]	-0.16 (0.043)	0.0	434	0.96	139
self-written plan text itself	0.80 [0.70, 0.89]	–	0.71 [0.60, 0.80]	–	0.0	–	–	–

some *other* secret: $h \leftarrow h - U_\ell^\top (U_\ell h - c_{w,\ell})$, with $c_{w,\ell} = U_\ell(\bar{h}_\ell - \frac{1}{14}\bar{z}_{w,\ell})$ estimated from set-A baseline stories. The prompt, which contains the secret, is processed normally. We use five conditions: the secret direction $\hat{d}_w$ (1 dimension per layer); the 14-d span of all 15 secret directions; and three controls, namely another word’s direction ($w+7$), a random direction, and a random 14-d subspace. Gemma-3 has massive activation dimensions; one coordinate (2339) holds $>99\%$ of the residual norm. We therefore zero the 8 largest-magnitude coordinates of every direction before normalising. Without this masking, projections are dominated by token-to-token fluctuations of the massive coordinates, and any clamp becomes a large perturbation (Section 4.6).

Steering (sufficiency). We generate set-B stories with *no* secret in the prompt and add $\alpha m_{24} \hat{d}_{w,24}$ at layer 24 on generated tokens, where m_{24} is the typical held-secret projection at that layer and $\alpha \in \{0, 4, 16\}$. We then ask whether the guesser can identify w .

Quality. Qwen3-14B rates each story’s fluency and coherence on a 1–9 scale (expected value over digit probabilities). We also report length and distinct-2.

4 Results

4.1 Gemma-3-12B leaks its secret word, and outlines stop it

Table 1 and Figure 1 (left) show the word-level results. The positive anchor replicates Holtzman and West [10]. With only the instruction to keep the secret, GPT-4.1 identifies the secret-holder 70% of the time (Qwen3-14B: 64%), and no story in this condition contains the secret word. The leak is involuntary in exactly the sense of the original paper: the writer complies with the letter of the instruction but not its spirit. On held-out premises (set B), the same condition gives 0.70 (Table 3).

An external plan removes the leak. When the writer follows a secret-free outline, guesser accuracy falls to chance. The short outline gives 0.54 (paired $\Delta = -0.16$, $p = 0.004$) and the detailed outline 0.48 ($\Delta = -0.22$, $p = 0.002$). Both guessers agree. The effect is not explained by **dilution**. Length-matched irrelevant text, which is *longer* than the short outline, leaves leakage unchanged (0.71, $\Delta = +0.01$), and the outline–irrelevant contrasts are significant (detailed: -0.23 , $p = 0.002$; short: -0.16 , $p = 0.014$). Short and detailed outlines do not differ significantly ($p = 0.18$), so even a five-bullet plan suffices. The plan’s effect is comparable to that of the decoy instruction, a known mitigation (0.50).

A plan written by the secret-holder does not help. When the model writes its own outline before the story, the outline itself leaks strongly (0.80). For *cactus*, for example, the plan begins with a “small, dusty hometown”. The story that follows leaks at least as much as the baseline (0.76, $\Delta = +0.06$, n.s.). What matters, then, is not planning as such but *who decides the content*. An outline made without access to the secret commits the content in advance. A plan made by the secret-holder is just more text into which the secret leaks.

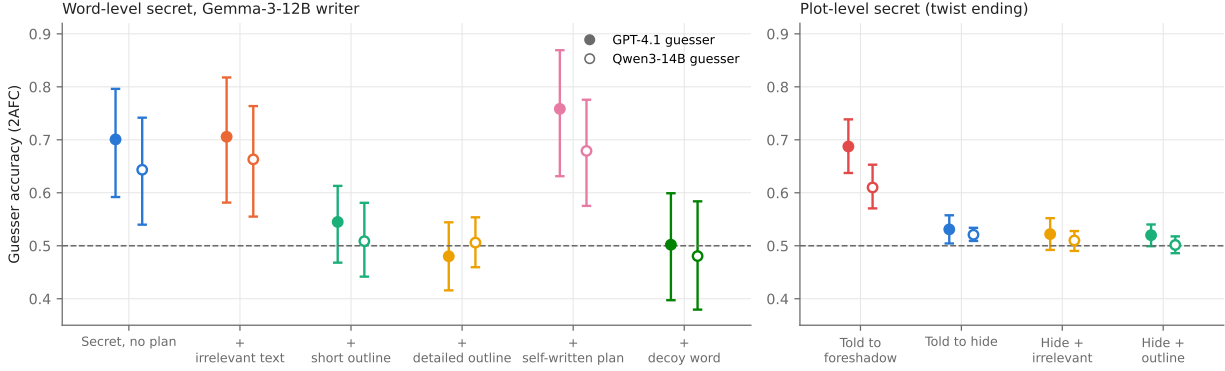


Figure 1: Guesser accuracy (95% crossed-bootstrap CIs) for word-level conditions (left) and plot-level twist secrets (right). Filled markers: GPT-4.1 guesser; hollow: Qwen3-14B guesser. Chance is 0.5.

Table 2: Plot-level (twist) leakage: 50 premises $\times$ 2 twists $\times$ 2 samples = 200 openings per condition. CIs from a premise-cluster bootstrap. Δ is the paired difference from *told to hide*.

Condition	GPT-4.1		Qwen3-14B	
	acc. [95% CI]	Δ vs. hide (p)	acc. [95% CI]	Δ (p)
Told to foreshadow (positive control)	0.69 [0.64, 0.74]	+0.16 (<.001)	0.61 [0.57, 0.65]	+0.09 (<.001)
Told to hide	0.53 [0.50, 0.56]	—	0.52 [0.51, 0.53]	—
Hide + irrelevant text	0.52 [0.49, 0.55]	-0.01 (0.678)	0.51 [0.49, 0.53]	-0.01 (0.383)
Hide + outline	0.52 [0.50, 0.54]	-0.01 (0.586)	0.50 [0.49, 0.52]	-0.02 (0.028)
No twist given: P(pick twist A)	0.51	—	0.59	—

4.2 Plot-level secrets leak only slightly

Table 2 and Figure 1 (right) show the twist results. The paradigm can detect foreshadowing: when asked to foreshadow, openings point to the right ending 69% of the time (GPT-4.1). When told to hide the twist, Gemma’s openings are much harder to read: 0.53 [0.50, 0.56] for GPT-4.1 and 0.52 [0.51, 0.53] for Qwen3, so the leak is small but its interval excludes chance for Qwen3. Adding the outline gives 0.52 (GPT-4.1) and 0.50 (Qwen3; paired $\Delta = -0.02$, $p = 0.03$), and irrelevant text gives 0.52 and 0.51. The no-twist control confirms that the premise alone does not reveal the ending once both twists are balanced. In short, plot-level secrets kept under an explicit “do not foreshadow” instruction leak far less than secret words do, and the remaining leak is at the edge of what our guessers can detect. The outline effect points in the predicted direction but is small in absolute terms. A plausible reason is that a twist is a complex proposition that the opening rarely needs to approach, whereas a single concrete word can be “spent” in any sentence.

4.3 The secret is held in the residual stream throughout the story

Figure 2 summarises the counterfactual teacher-forcing analysis. Four results stand out.

(1) **The held secret is linearly decodable from story-token activations even with the text fixed.** In baseline stories, 15-way decoding of the prompt secret from the mean story-token residual reaches 0.81 at layer 24 and 0.97–1.00 from layer 28 to the output (chance 0.067), using directions estimated on other premises. Since the text is identical across the 15 runs, this is information the model *carries*, not information in the words. It is present in every tenth of the story (Figure 3).

(2) **The model “thinks the word” but suppresses it at the output.** With the logit lens restricted to the 15 candidate words and centred across prompt variants, the held secret is the top candidate in 79–97% of baseline stories at layers 8–40. At the final layer it is top in only 1%. There the secret’s logit is *lower* than under other prompts (Figure 3, right), reflecting compliance with “do not mention”. This is the white-bear pattern [13, 19]: suppressing a concept requires representing it.

(3) **The secret grows more active as the story goes on, and the output suppression weakens.** In the baseline and

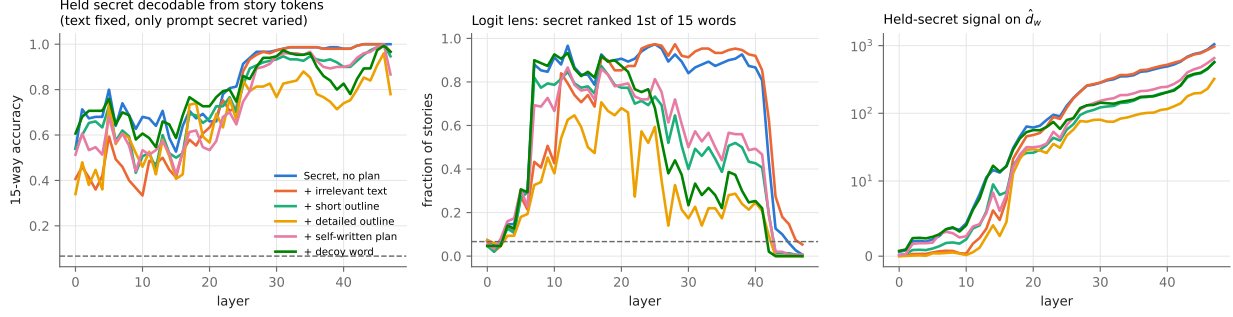


Figure 2: Counterfactual activation analysis (Gemma-3-12B; 150 stories per condition). Left: 15-way decoding of the held secret from mean story-token residuals with the text fixed. Middle: fraction of stories where the logit lens ranks the secret first among the 15 candidate words (centred across prompt variants). Right: held-secret signal (projection on $\hat{d}_w$ minus mean projection on other directions; symlog scale). Dashed lines: chance.

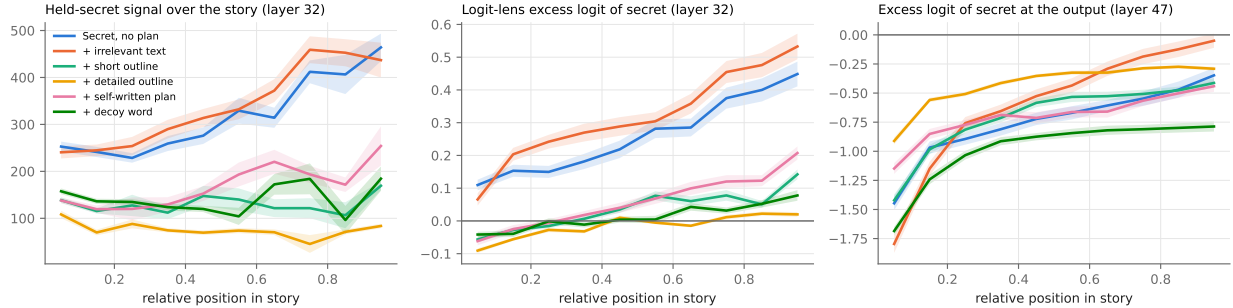


Figure 3: Time course within the story (10 bins; shaded: ± 1 s.e.). In the leaking conditions (secret, irrelevant text), the held secret becomes *more* active as the story proceeds and output-layer suppression relaxes. Under outlines and decoys it stays flat.

irrelevant-text conditions, the held-secret signal at layer 32 nearly doubles from the first to the last tenth of the story (253→464). The logit-lens excess at layer 32 rises from 0.11 to 0.45, and the output-layer suppression relaxes from -1.45 to -0.35 . Since the text is held fixed across variants, this signal can include interactions between the held secret and the story content. Section 4.4 shows, however, that the rise is not mainly driven by leaked content.

(4) Internal signal predicts leakage. Across stories, the per-story held-secret signal correlates with that story’s guesser accuracy: Spearman $\rho = 0.46$ at layer 32 in the baseline ($p < 10^{-8}$) and 0.61 with irrelevant text. Pooling all conditions and ranking within each condition gives $\rho = 0.28$ ($p < 10^{-10}$; Figure 4). The correlation emerges at the same layers (≥ 24) where decoding saturates. Across conditions (Figure 4, right), mean internal signal and leakage line up. Baseline and irrelevant text have high signal and high leakage. Short outline (signal 128), decoy (139), and detailed outline (75) have low signal and chance leakage. The one exception is the self-written plan (signal 167, leakage 0.76), where the secret has already been written into the plan text and the story can leak by continuing that text.

Outlines weaken the representation without erasing it. Under the detailed outline the secret is still decodable (0.83 at layer 32, versus 0.99), but its signal is about $4\times$ smaller and does not grow over the story. The logit-lens top-1 rate at layers 28–40 drops from 0.84–0.93 to 0.14–0.33, and output-layer suppression is weaker from the start. Length-matched irrelevant text changes none of these quantities. The outline therefore does not merely compete for attention. Once the content is fixed, the model draws on the secret much less while writing, yet it still knows the secret.

4.4 Is the weaker signal caused by the plan in context or by the text it produced?

Outline-conditioned stories differ from baseline stories in two ways: the outline is in the context, and the text itself is different, with no leaked content to resonate with. To separate the two, we re-ran the counterfactual analysis with text and context swapped (Figure 5). *Outline-condition stories re-scored without the outline in context* carry a full-strength secret: signal 304 at layer 32 (baseline 318), decoding 1.00, logit-lens top-1 0.93, and the same rise over the story.

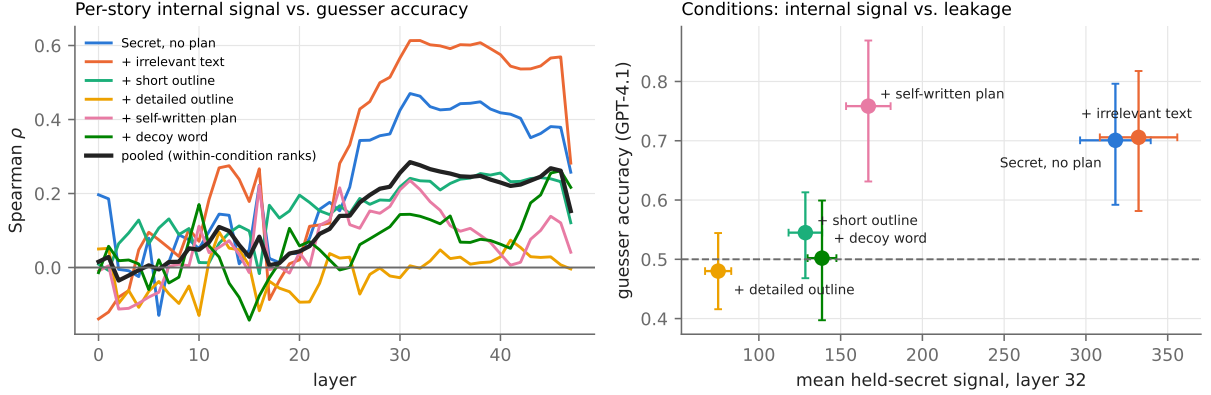


Figure 4: Left: Spearman correlation, by layer, between a story’s held-secret signal and its guesser accuracy (GPT-4.1); black: pooled over conditions using within-condition ranks. Right: condition means of signal (layer 32) and leakage. The self-written plan is the outlier because its secret is already in the plan text.

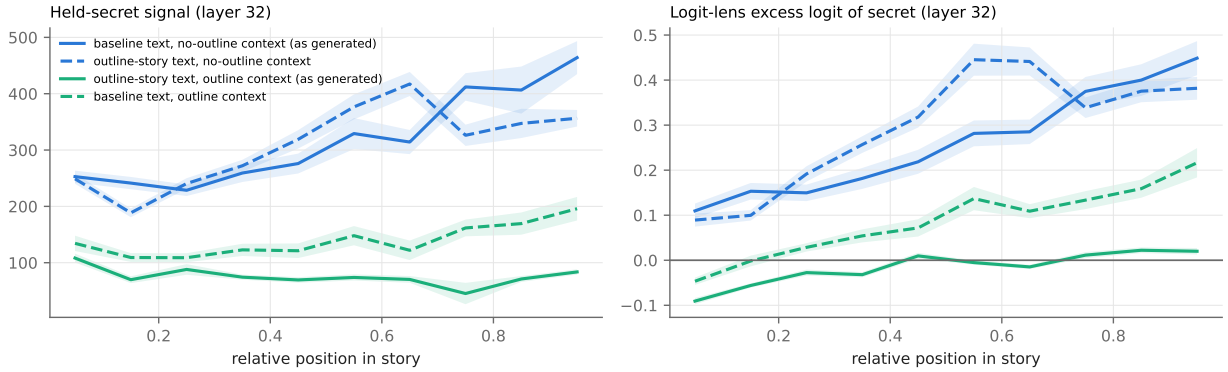


Figure 5: Cross-context test. Solid lines: stories scored in the context in which they were generated. Dashed lines: the same texts re-scored with the other context. Colour indicates the *context* (blue: no outline; green: outline). The held-secret signal follows the context, not the text.

Baseline stories re-scored with the outline in context carry a much weaker secret: signal 137, decoding 0.93, top-1 0.63. Having the plan in context is therefore what quiets the secret. The text accounts for a smaller remainder: leaky baseline text still gets more signal than outline text under the same outline context (137 vs. 75). The rise over the story also appears for the non-leaky outline text without the outline in context, so it is not mainly a feedback loop from leaked content. It looks instead like the writer increasingly consulting the secret whenever it must decide content on its own.

4.5 Interventions: removing the secret direction reduces leakage, and adding it creates leakage

Table 3 and Figure 6 report interventions during generation on the held-out premises. Mean-ablating the secret direction ($\hat{d}_w$, one dimension per layer) lowers leakage from 0.70 to 0.59 under GPT-4.1 ($\Delta = -0.12$, $p = 0.075$), and from 0.62 to 0.56 under Qwen3 with both samples pooled ($\Delta = -0.06$, $p = 0.17$). Ablating the 14-d subspace spanned by all secret directions lowers it to 0.57 (GPT-4.1, $\Delta = -0.13$, $p = 0.045$) and 0.54 (Qwen3 pooled, $\Delta = -0.09$, $p = 0.04$). The matched controls (another word’s direction, a random direction, a random 14-d subspace) change leakage by between -0.01 and $+0.03$, all far from significant. Directly against their matched controls, the secret subspace beats the random subspace under both guessers (GPT-4.1 -0.13 , $p = 0.04$; Qwen3 pooled -0.10 , $p = 0.02$). The single secret direction beats the other-word direction by -0.10 (GPT-4.1) and -0.07 (Qwen3), with $p \approx 0.10$ in both cases. Ablation does not degrade the stories: judge ratings are 8.51–8.59 against 8.59 without ablation, and length, distinct-2

Table 3: Mean-ablation at every layer on generated tokens (held-out set-B premises). Δ is the paired difference from no ablation, with crossed-bootstrap p . GPT-4.1 scored the first sample of 150 stories per condition. Qwen3-14B scored both samples (two independent seeds, 300 stories) where available. Quality is a Qwen3-14B judge rating plus length.

Mean-ablated subspace	GPT-4.1 (sample 1)		Qwen3-14B (samples pooled)		n	quality (sample 1)	
	acc. [95% CI]	Δ (p)	acc. [95% CI]	Δ (p)		judge (1–9)	words
No ablation	0.70 [0.60, 0.81]	–	0.62 [0.54, 0.70]	–	300	8.59	427
Secret direction $\hat{d}_w$ (1-d)	0.59 [0.50, 0.68]	-0.12 (0.075)	0.56 [0.49, 0.62]	-0.06 (0.171)	300	8.58	432
Secret subspace (14-d)	0.57 [0.50, 0.65]	-0.13 (0.045)	0.53 [0.47, 0.60]	-0.09 (0.040)	300	8.51	427
Other word’s direction (1-d)	0.69 [0.58, 0.79]	-0.01 (0.806)	0.62 [0.54, 0.71]	+0.00 (0.954)	300	8.57	429
Random direction (1-d)	0.70 [0.59, 0.80]	-0.00 (0.936)	0.65 [0.56, 0.74]	+0.02 (0.658)	150	8.59	430
Random subspace (14-d)	0.70 [0.60, 0.80]	-0.01 (0.883)	0.64 [0.54, 0.74]	+0.02 (0.735)	300	8.46	424

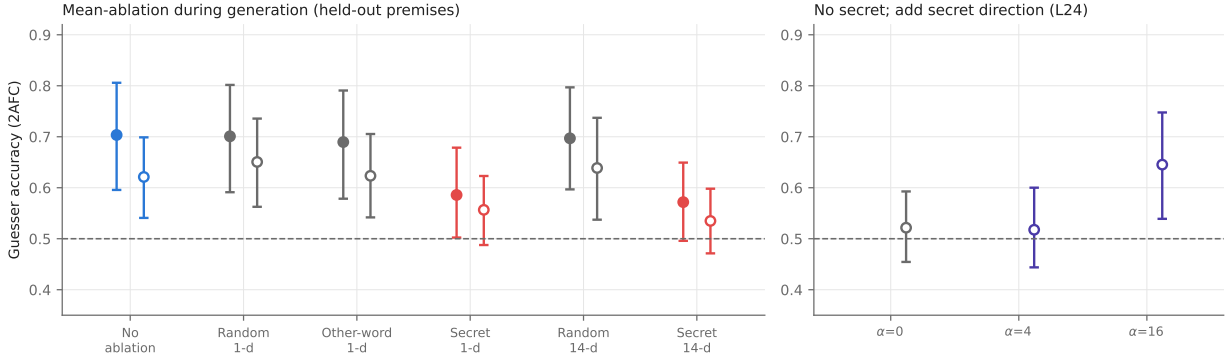


Figure 6: Left: leakage under mean-ablation (filled: GPT-4.1, sample 1; hollow: Qwen3-14B, pooled samples). Right: steering stories written without a secret by adding $\alpha m \hat{d}_w$ at layer 24 (Qwen3-14B guesser).

and the absence of explicit mentions are unchanged. Removing the linear secret representation therefore removes a specific part of the leak, about half of the above-chance accuracy, but not all of it. This is expected, since the secret remains readable from the prompt at every layer and may also be encoded nonlinearly.

Adding the secret direction to stories written *without* any secret is enough to induce detectable leakage when the push is large. With $\alpha = 16$ ($16\times$ the typical held-secret projection, at layer 24 only), Qwen3 identifies the injected word 0.65 [0.54, 0.75] of the time, and 0.62 [0.52, 0.72] after excluding questions where a story mentions the word. At $\alpha = 4$ accuracy is 0.52, the same as with no injection ($\alpha = 0$: 0.52). The required push being large reflects that natural secret-holding is spread over all layers and tokens, whereas we inject at one layer. Steering does not hurt quality (judge 8.71 vs. 8.76). Steering was scored only by Qwen3-14B, because the API budget was exhausted.

4.6 A pitfall: zero-ablation amplifies the concept

Our first intervention projected $\hat{d}_w$ out entirely, setting $h \cdot \hat{d}_w = 0$. The resulting “stories” were degenerate repetitions of the secret’s neighbourhood: *lighthouse* $\rightarrow$ “The coastal coastal coastal...”, *violin* $\rightarrow$ “TheMusicMusicmusic...”, *cactus* $\rightarrow$ “The desert [cactus emoji] desert desert...”, *copper* $\rightarrow$ “The copper copper...”. In all, 20/150 stories contained the word itself. The cause is Gemma-3’s massive activations. One residual coordinate carries $>99\%$ of the norm ($|h_{2339}| \approx 7 \times 10^4$ at layer 24), so the uncentred projection on $\hat{d}_w$ is around $\pm 10^4$, two orders of magnitude larger than the secret-induced shift ($\approx 10^2$). Zeroing it therefore applies a huge push along $\hat{d}_w$, which acts as steering towards the concept rather than ablating it. Even mean-clamping fails unless the massive coordinates are masked out of the direction, because their token-to-token fluctuations dominate the projection. Directional-ablation results on models with massive activations should be read with this in mind.

Table 4: Llama-3.3-70B writer (150 stories per condition, GPT-4.1 guesser).

Llama-3.3-70B writer	GPT-4.1 acc. [95% CI]	Δ vs. secret (p)	words
Secret, no plan	0.58 [0.50, 0.65]	—	447
+ irrelevant text	0.53 [0.45, 0.62]	-0.05 (0.309)	421
+ short outline	0.48 [0.41, 0.56]	-0.10 (0.088)	456
+ detailed outline	0.52 [0.46, 0.58]	-0.06 (0.244)	637

4.7 Cross-model check: Llama-3.3-70B

We repeated the plan manipulation with Llama-3.3-70B-Instruct as the writer (via API; its outlines also written secret-free by Llama itself), with the GPT-4.1 guesser (Table 4). In our setup Llama leaks only weakly (0.58 [0.51, 0.65]), unlike the 83–84% reported by Holtzman and West [10], possibly because of prompt and guesser differences. All plan and control conditions fall near chance (0.48–0.53), and no contrast is significant, so this replication is underpowered and does not distinguish plan from dilution.

5 Discussion

What the plan does. The hypothesis behind this study was that leakage comes from entanglement between planning and prose generation. A model holding a secret has to keep deciding what comes next, and those decisions are biased by what it is holding in mind. Our results support this account, with two refinements. First, conditioning on a plan removes leakage, but only when the plan was made *without* the secret. A self-made plan is just more generation and leaks like any other text, so planning helps by moving content decisions out of the secret-holder’s hands. Second, the plan does not erase the secret. It is still decodable at 83–93% (15-way) from story tokens, but it is used much less. Without an outline, the secret becomes *more* active as the story goes on, and output suppression relaxes. With one, the secret signal stays low and flat, and the cross-context test (Section 4.4) shows that it is the plan in context, not the resulting text, that quiets it. This fits the picture of a writer who, when it must invent the next beat, keeps consulting what it knows. When the beat is given, it has no reason to.

Mechanism of the leak. The leak is mediated, at least in part, by a representation that is linearly accessible in the residual stream at story positions. Its strength predicts per-story leakage; removing one direction per layer reduces leakage (by about 0.12 for GPT-4.1, with a smaller effect under Qwen3, Table 3) at no cost to quality; and adding the direction to a no-secret model induces leakage. The ablation removes only part of the leak. This is expected, because the secret is also available by attention to the prompt at every layer, and much of its representation is likely nonlinear or spread across many directions. The logit lens adds a cognitive-style reading: the model internally ranks the forbidden word first and suppresses it only in the last few layers, the transformer analogue of ironic rebound. Leakage then happens through *associated* content (rain, dust, coastlines) that the final-layer suppression does not target, which matches the semantic-leakage phenomenon of Gonen et al. [9].

Practical implications. For writing systems that must not telegraph information (surprise endings, hidden-role games, red-team “sleeper” personas, or simply not revealing private context), the simplest effective mitigation we found is to have the content decided by a process that does not see the secret, and to condition generation on that plan. Merely adding context does not help, and asking the secret-holder to plan first can make things worse. In our data the decoy instruction works equally well, but it requires choosing a concept to leak instead.

6 Limitations

- **One main writer.** Mechanistic results are for Gemma-3-12B-it only. The Llama-3.3-70B replication leaked too weakly in our setup to test the plan effect (Table 4).
- **Guessers.** The OpenRouter key had a shared daily budget, so GPT-4.1 judgements cover all main conditions but not the steering runs or the second ablation sample, which only the local Qwen3-14B scored. Qwen3 is less

sensitive (baseline 0.64 vs. 0.70), but the two guessers agree qualitatively everywhere. Guessers are strongly position-biased, and we cancel this by averaging both orders.

- **Statistical power.** With 150 questions per condition and large word-to-word heterogeneity, the crossed bootstrap gives wide intervals (± 0.1). The ablation effects are consistent across guessers but individually only marginally significant.
- **Outlines change the stories.** Outline-conditioned stories follow the outline closely and are therefore more similar to each other within a pair. One might worry that this hides leakage. Because both stories in a pair share the same outline, however, any secret-driven difference should be *easier* to spot. Detailed-outline stories are also longer (530 vs. 433 words), and some reached the 800-token limit.
- **Counterfactual signal includes text–secret interactions.** With the text fixed and the prompt varied, z also captures how the held secret modulates the processing of whatever content is present. The cross-context test (Section 4.4) suggests this contribution is secondary. Directions are difference-of-means estimates from one condition, and we did not use SAEs.
- **Plot-level materials** were generated by an LLM (Claude Sonnet 4.5). The twist pairs are not perfectly balanced in salience (the Qwen3 guesser favours twist A 59% of the time with no twist given), although the balanced design cancels this in the accuracy metric.

7 Conclusion

LLMs are indeed bad at keeping obvious secrets. Gemma-3-12B keeps the forbidden word internally active while writing, and that activity, which grows as the story goes on, leaks into the prose as associated content. The original question was whether conditioning on a plan that sets up the future would help. It does: an outline written without the secret brings leakage to chance, length-matched irrelevant context does not, and a plan written by the secret-holder leaks as well. The plan does not make the model forget the secret, but it largely stops the model from consulting it. Removing the secret direction from the residual stream reduces leakage, and injecting it creates leakage. Twist endings kept under an explicit instruction leak only marginally, and an outline brings them to chance as well.

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A Prompts

Word-level (secret, no plan). *Your secret word is “{w}”. Keep it secret: do not mention the word and do not hint at it in any way. The story must not reveal anything about the secret word.* \n\n *Write a short story (about 450 words) about {premise}.* \n\n *Output only the story text, with no title.* Outline conditions insert *Follow this outline closely, beat by beat:* plus the outline before the last line. The irrelevant condition inserts *Below is an unrelated reference passage that is part of this prompt template. It has nothing to do with your story; ignore it.* plus the passage. The self-plan condition replaces the last line with *First write a brief outline of the story under the heading OUTLINE (5-8 bullet points). Then write the full story under the heading STORY.* The decoy condition appends *Instead, think about the word “{d}” while you write.* to the secret line.

Outline generation (no secret). *Write a detailed beat-by-beat outline (12 numbered beats, about 300 words in total) for a short story (about 450 words) about {premise}. Each beat should say concretely what happens, including the key images, objects and dialogue moments.* The short variant asks for 5 bullet points and about 80 words.

Plot-level. *Here is the premise of a short story: $\{premise\}$. The story will end with this twist, revealed only in the final scene: $\{twist\}$. Write the *OPENING* of the story (about 400 words). Only write the opening; the rest of the story, including the ending, will be written later in a separate part that you will not write. Do not reveal, hint at, or foreshadow the twist in any way: a reader of the opening must not be able to guess how the story ends.*

Guesser (word-level). *Two short stories were written in response to the same writing prompt. Each writer was privately given a different secret word and instructed not to reveal it. Exactly one of the two writers had the secret word “ $\{w\}$ ”. Which story was written by the writer whose secret word was “ $\{w\}$ ”? Look for any subtle traces: imagery, themes, objects, word choice or mood that might relate to “ $\{w\}$ ”. [stories] Answer with a single digit: 1 or 2.*

B Additional details

Pairing. For premise p , secret i is paired with secret $j = (i + 1 + (p \bmod 14)) \bmod 15$, so each story serves once as the target and once as the foil, and pairings vary across premises.

Directions and masking. Secret directions are difference-of-means vectors computed separately for each of the 48 layers. For interventions, we zero the 8 coordinates with largest mean $|h|$ in each layer (including the massive coordinate 2339) before normalising. The steering scale $m_{24} = 74.4$ is the mean held-secret projection at layer 24 in baseline stories.

Compute. One RTX A6000 GPU. Generation: about 3 GPU-hours. Activation analysis: about 2 GPU-hours. Local guessing and quality judging: about 1 GPU-hour. API cost was under \$30.